

An MLIR-Based Approach to HPC Portability

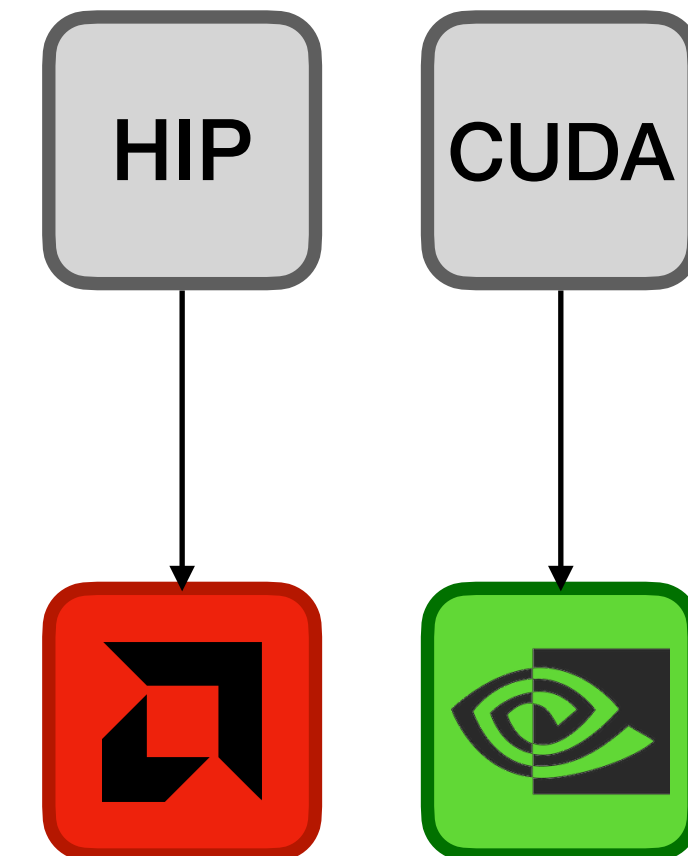
Mojo

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Motivation

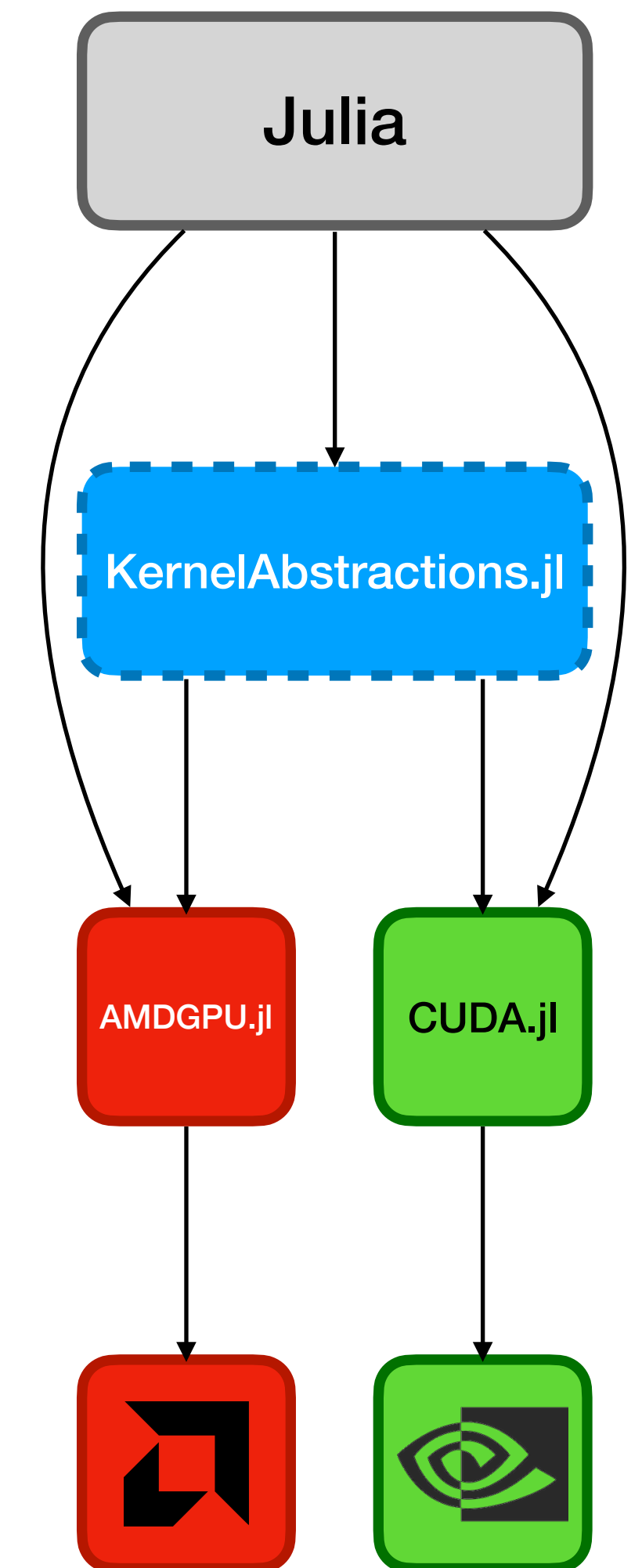
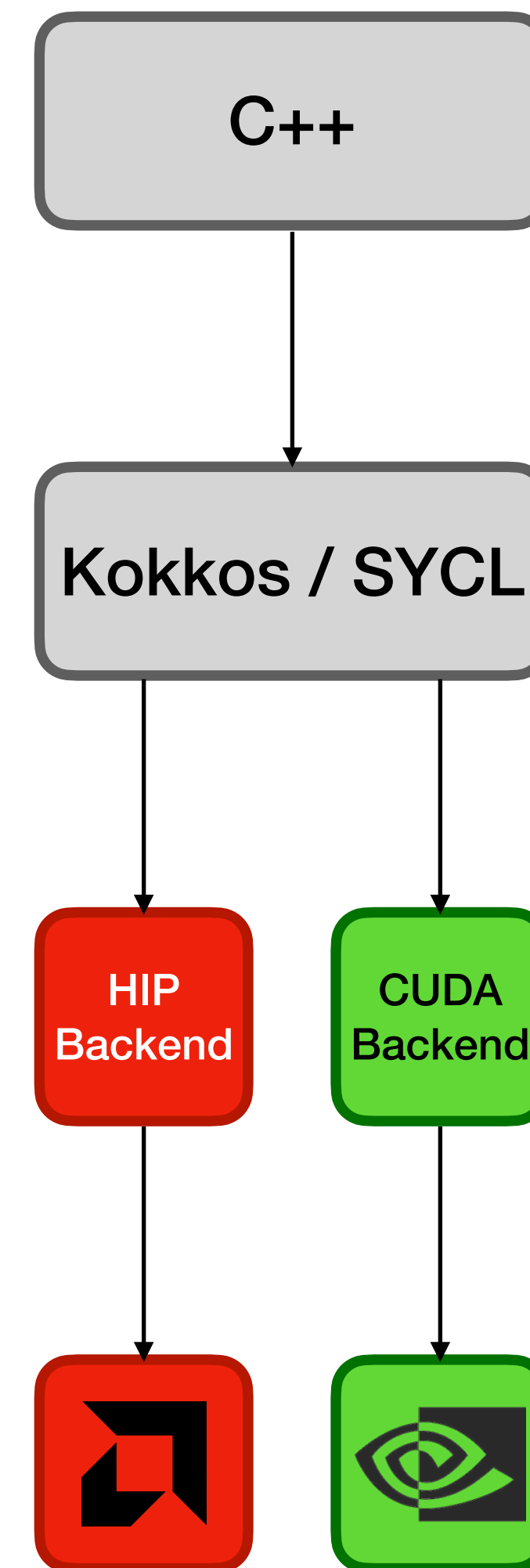
How can we write GPU Kernels?

- Naive approach to writing a NVIDIA/AMD kernel.
 - Just use CUDA/HIP
- Leads to some issues
 - Vendor lock-in
 - Language Split



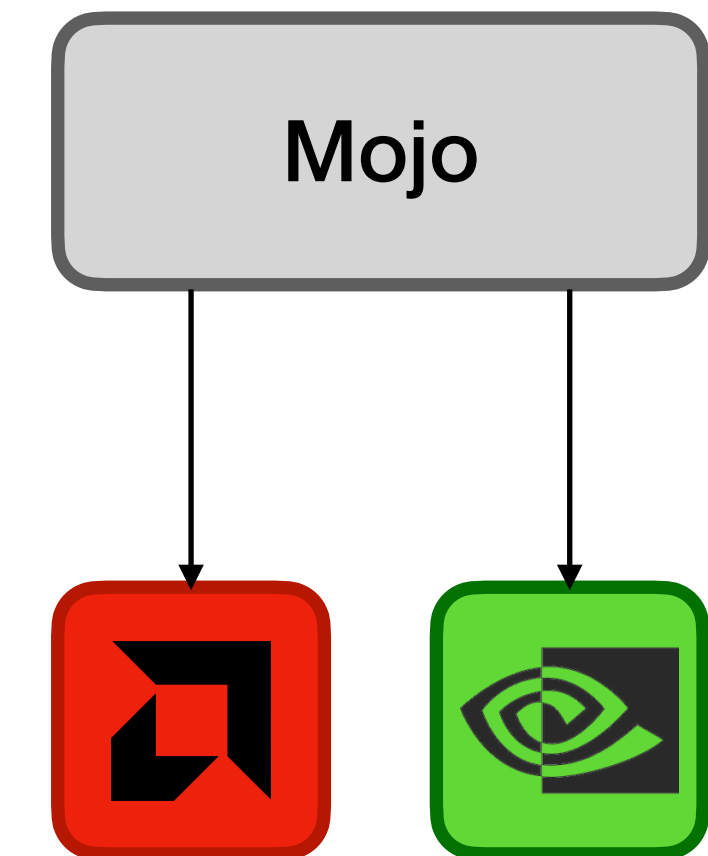
How can we improve on this?

- Addressing Vendor lock in
 - Kokkos/SYCL, production proven
 - One joined backend for generating both NVIDIA and AMD kernels
- Addressing language split
 - Julia, JIT to near-native, mature ecosystem
 - One language to program kernels and prototype



Can we address both issues?

- Mojo attempts addressing both issues
- Superset* of python prototyping and kernel development are both pythonic
 - Python interop runs through CPython at runtime
- Use the compiler to generate both NVIDIA and AMD backends



Background

LLVM

- One shared IR for all languages
 - One set of optimizations for C, C++, Rust, Swift, ...
- Compiler loses significant information
 - Layout, access pattern, other semantics
- To solve issues languages/frameworks often develop own compiler stacks with many mid level IRs

```
for (int i = 0; i < N; i++)  
  for (int j = 0; j < N; j++)  
    for (int k = 0; k < N; k++)  
      C[i][j] += A[i][k] * B[k][j];
```

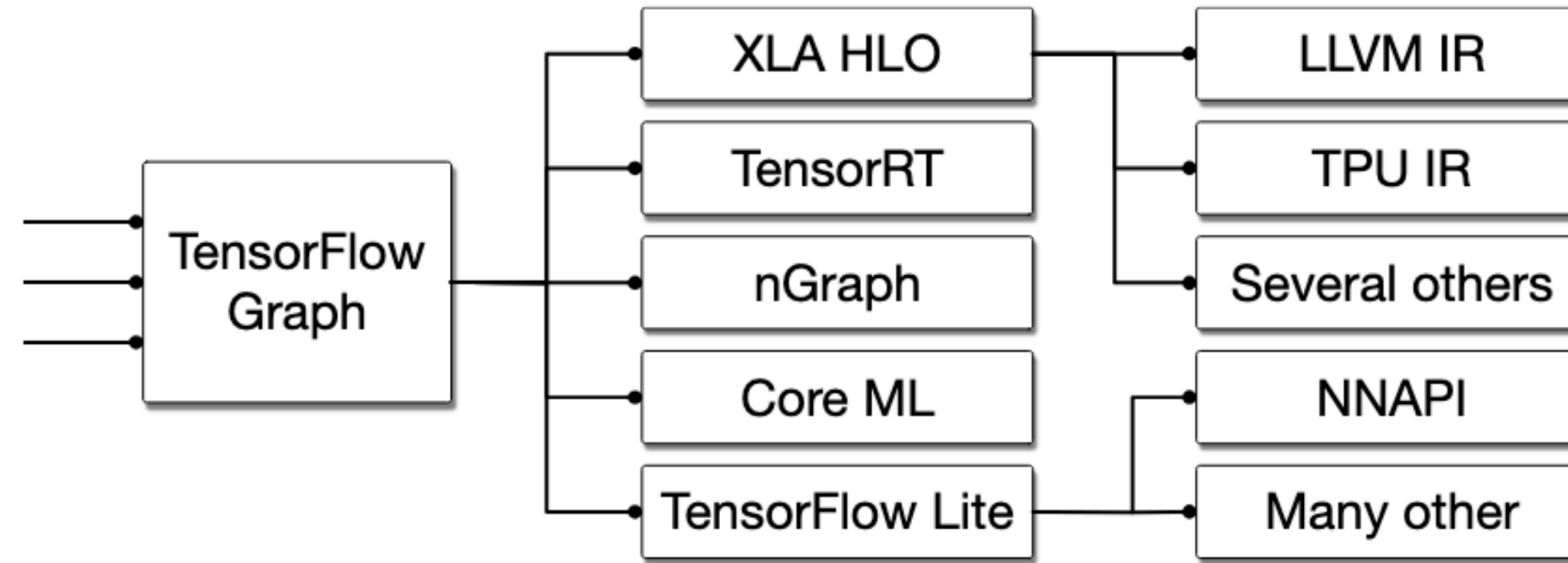
Next slide...

```
%a_p = getelementptr float,float* %A,i64 %a_i  
%a_v = load float, float* %a_p  
%b_p = getelementptr float,float* %B,i64 %b_i  
%b_v = load float, float* %b_p  
%prod = fmul float %a_v, %b_v  
%c_p = getelementptr float,float* %C,i64 %c_i  
%c_v = load float, float* %c_p  
%sum = fadd float %c_v, %prod  
store float %sum, float* %c_p
```

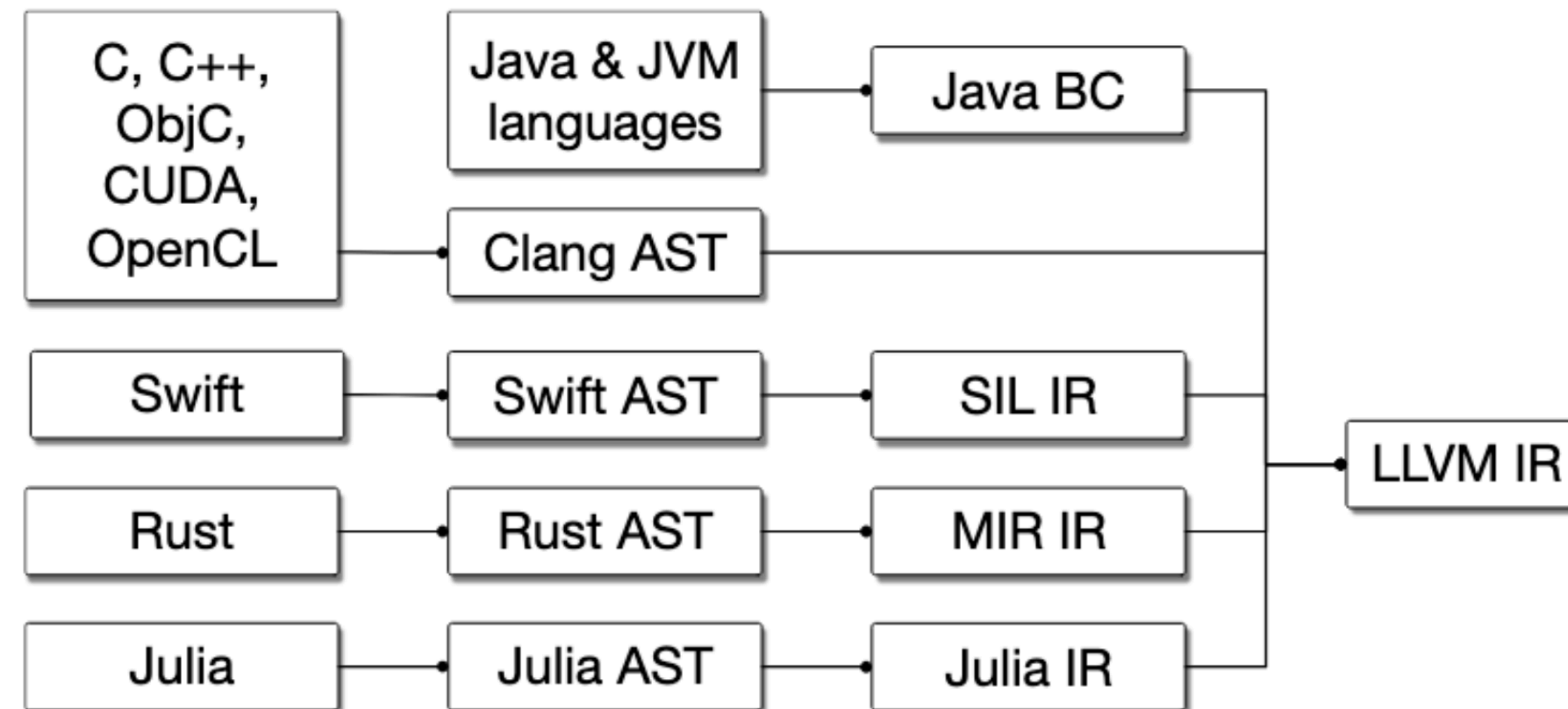
Excerpt

Example of mid level IRs

Frameworks:



Languages:

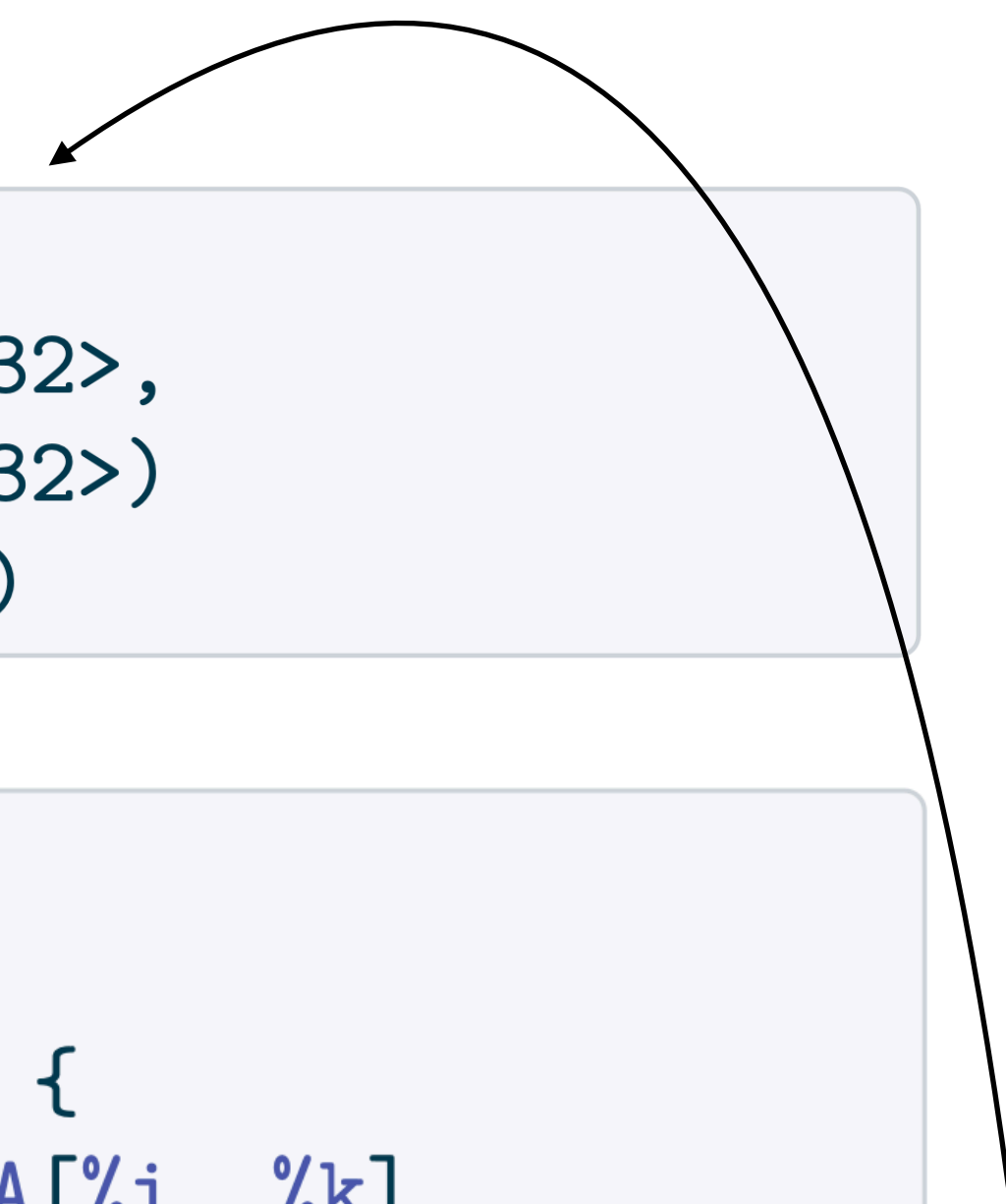


MLIR

- Dialects: namespaces sets of ops/types (linalg., affine., nvvm.)
- Progressive lowering, structure kept as long as it's useful
 - linalg -> affine -> scf -> LLVM IR
- Different parts of one program can sit at different levels at the same time, providing compiler info
- Different IRs are no longer needed, optimizations can be reused within same framework

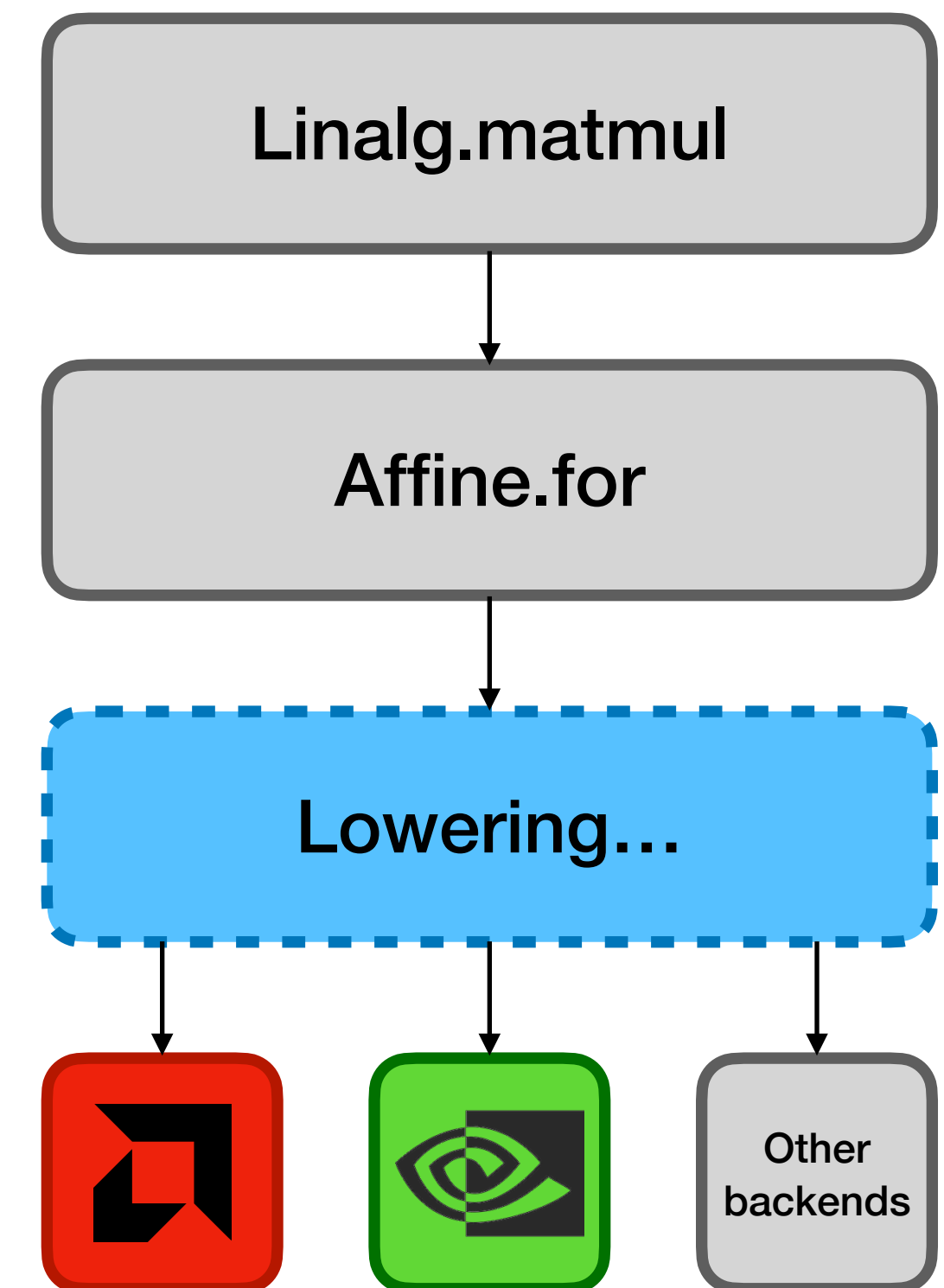
```
linalg.matmul
  ins(%A, %B : memref<?x?xf32>,
      memref<?x?xf32>)
  outs(%C : memref<?x?xf32>)
```

```
affine.for %i = 0 to %M {
  affine.for %j = 0 to %N {
    affine.for %k = 0 to %K {
      %a      = affine.load %A[%i, %k]
                : memref<?x?xf32>
      %b      = affine.load %B[%k, %j]
                : memref<?x?xf32>
      %c      = affine.load %C[%i, %j]
                : memref<?x?xf32>
      %prod   = arith.mulf %a, %b : f32
      %sum    = arith.addf %c, %prod : f32
      affine.store %sum, %C[%i, %j]
                : memref<?x?xf32>
    } } }
```



Payoff

- Portability through dialects
- Same high-level IR + a different set of lowering passes go to a different hardware target
- Hardware-specific code exists only in the final lowering step
 - portability is a property of the compiler infrastructure



Concept

Mojo

- The first language built as MLIR dialects
- Superset of Python, adding exactly what the compiler needs
 - static typing
 - ownership
 - compile-time metaprogramming

Solving our problems

- Two language problem
 - numpy/scipy import directly, no FFI
 - However interop is runtime-only
- Vendor lock-in
 - New hardware target = one new lowering dialect. Portability is in the compiler.

```
def add(x, y):  
    return x + y  
  
def add(x: Float32, y: Float32) -> Float32:  
    return x + y
```

Code

- parameters are resolved at compile time
- structs and TileTensor have fully static layout, MLIR sees the memory access pattern
- problem shape must be fixed at compile time
 - awkward for adaptive meshes or anything dynamically shaped
- Same def, same language*

```
...

comptime dtype      = DType.float32
comptime N          = 1024
comptime tile       = 16
comptime layout     = row_major[N * N]()
comptime num_tiles  = ceildiv(N, tile)

def matmul_kernel(
    C: TileTensor[dtype, type_of(layout),
                  MutAnyOrigin],
    A: TileTensor[dtype, type_of(layout),
                  MutAnyOrigin],
    B: TileTensor[dtype, type_of(layout),
                  MutAnyOrigin],
):

    var row = block_idx.y * block_dim.y
              + thread_idx.y
    var col = block_idx.x * block_dim.x
              + thread_idx.x
    if row < N and col < N:
        var acc = Scalar[dtype](0)
        for k in range(N):
            acc += A[row * N + k]

                    * B[k * N + col]
            C[row * N + col] = acc
```


Evaluation

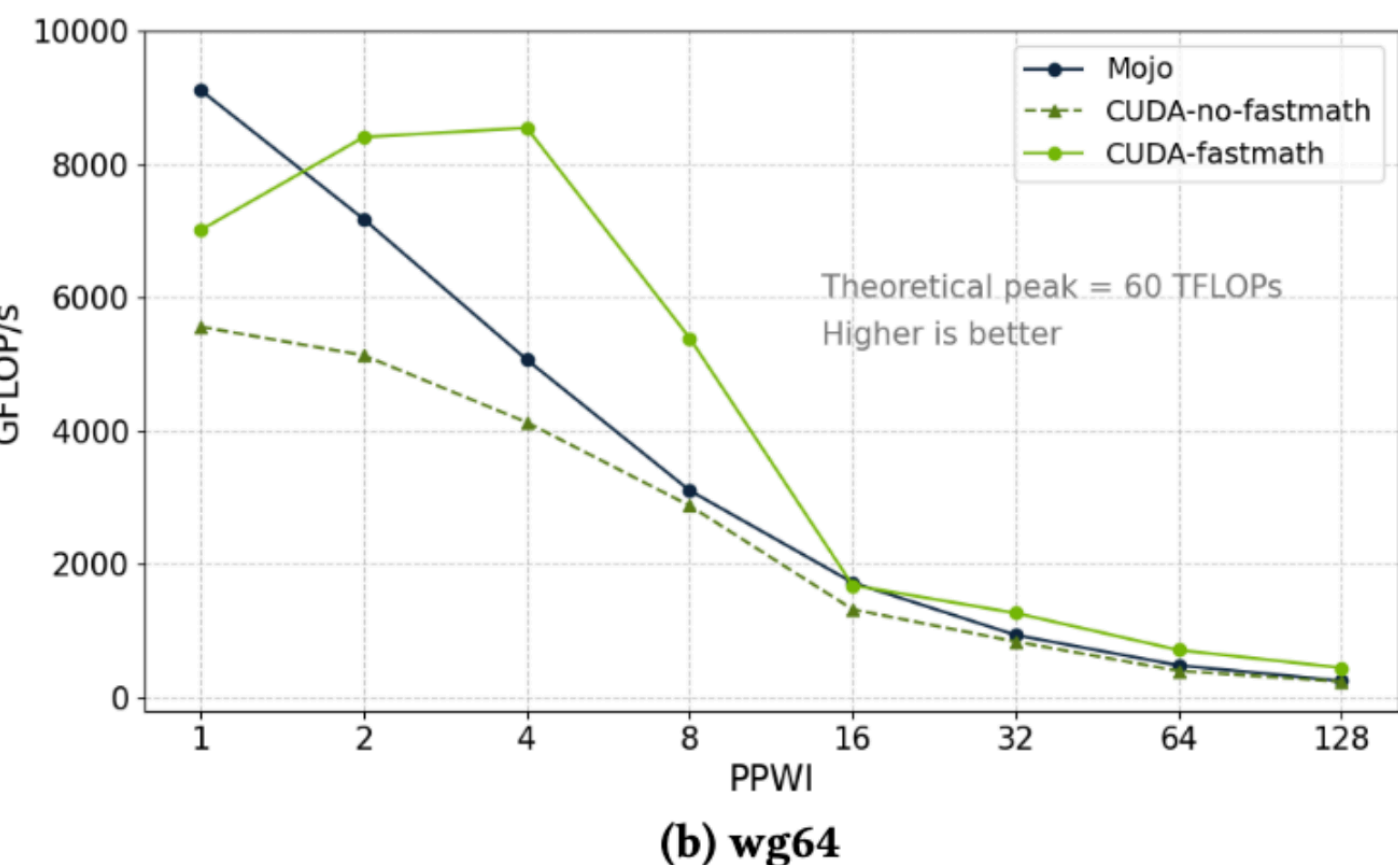
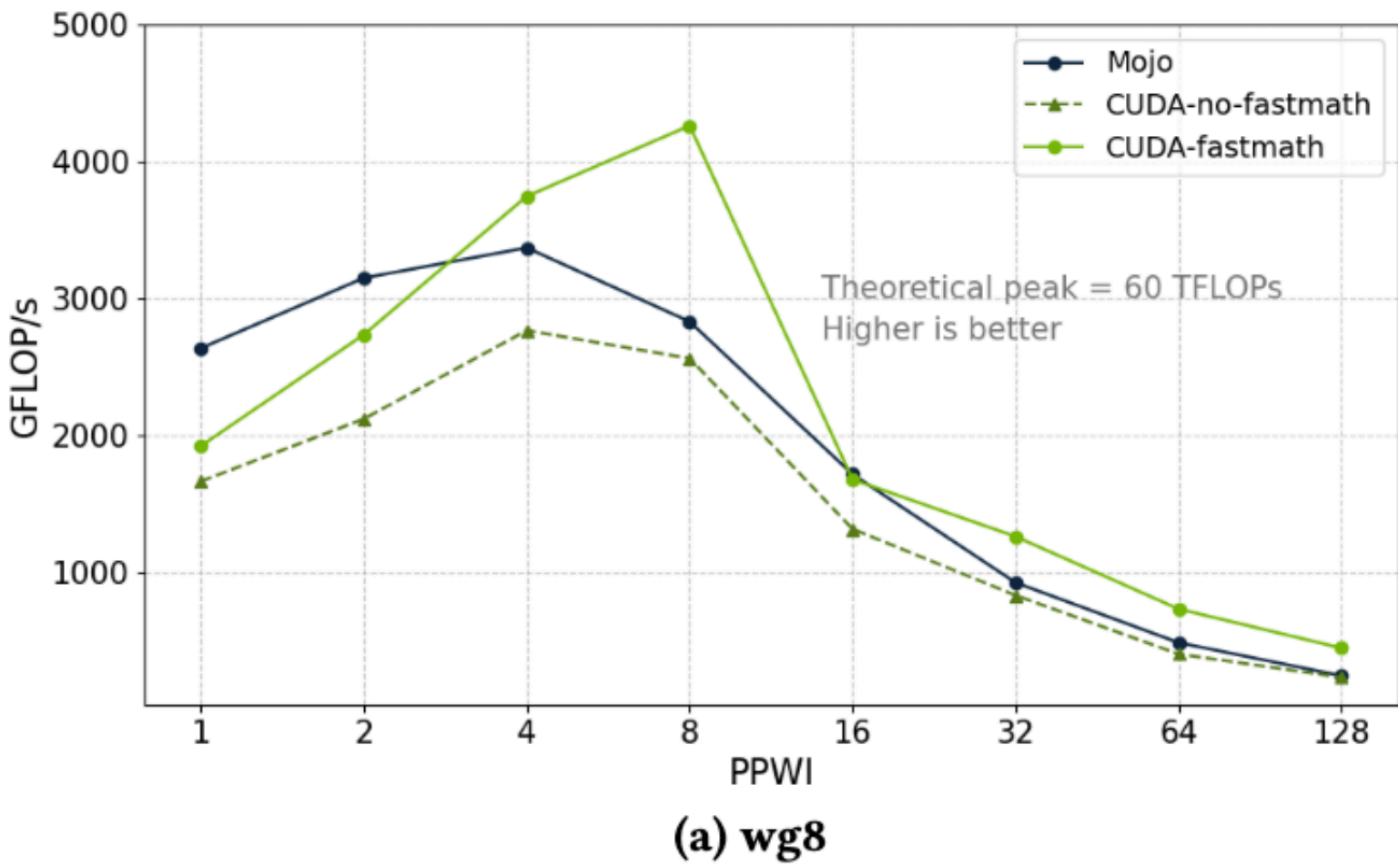
Setup

- Benchmarks from Godoy et al. (Oak Ridge National Lab) we survey their results
- Four HPC proxy kernels, Mojo vs. vendor baselines
 - CUDA on NVIDIA H100 and HIP on AMD MI300A
- memory-bound (bottleneck: moving data)
 - seven-point stencil, BabelStream
- compute-bound (bottleneck: arithmetic)
 - miniBUDE, Hartree-Fock
- Evaluate with Φ metric (T is set of GPU platforms)

$$\Phi = \frac{\sum_{i \in T} e_i}{|T|}, \quad e_i = \frac{\text{Mojo}_{\text{perf},i}}{\text{vendor}_{\text{perf},i}}$$

Results

- Memory bound
 - Seven-point stencil, $\Phi = 0.92$
 - BabelStream, $\Phi = 0.96$
- Compute bound
 - miniBUDE $\Phi = 0.54$
 - Hartree-Fock $\Phi = 0.92$ (!)
- Φ collapses two platforms into one number. miniBUDE: $\Phi = (0.82 + 0.59 + 0.38 + 0.38) / 4 = 0.54$ (!)



Kernel execution duration (ms)	NVIDIA H100		AMD MI300A	
	Mojo	CUDA	Mojo	HIP
a=1024 ngauss=6	147,250	2,652	–	846
a=256 ngauss=3	187	472	25,266	178
a=128 ngauss=3	21	53	2,765	23
a=64 ngauss=3	3	7	436	4

Verdict

- memory-bound
 - near vendor parity on both platforms
- compute-bound
 - real shortfalls, missing fast-math, atomic-operation issues on AMD

Mojo Efficiency	NVIDIA H100	AMD MI300A
7-point stencil		
FP32	0.82	1.00
FP64	0.87	1.00
$\overline{\Phi} = 0.92$		
BabelStream		
Copy	1.01	1.00
Mul	1.02	1.00
Add	1.01	1.00
Triad	1.01	1.00
Dot	0.78	1.00
$\overline{\Phi} = 0.96$		
miniBUDE		
PPWI=8 wg=8	0.82	0.38
PPWI=4 wg=64	0.59	0.38
$\overline{\Phi} = 0.54$		
Hartree-Fock		
a=1024 ngauss=6	17E-3	–
a=256 ngauss=3	2.52	7E-3
a=128 ngauss=3	2.52	8E-3
a=64 ngauss=3	2.33	8E-3
$\overline{\Phi} = 0.92$		

Conclusions and outlook

Critique

- Parses like Python, writing means thinking like a systems programmer
 - explicit thread indexing, compile-time sizes, manual memory layouts
- Python's ergonomics are absent
- Interop is runtime-only. Mojo world and Python stay separate (Invisible costs GIL and Conversion...)
- Lock-in irony. Mojo not yet open source, Modular is the new vendor
- Compilation times can get heavy

```
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        for k in range(N):
            acc += A[row * N + k]

                    * B[k * N + col]
            C[row * N + col] = acc
```

Conclusions

- Julia still wins the language split
- Kokkos/SYCL still win portability
- Mojo today, ergonomics gap, no fast-math, single-node only, missing portable libraries, incomplete AMD tooling, pre-1.0
- Despite gaps, only approach whose architecture targets both problems at once, gaps seem to be maturity/implementation holes.

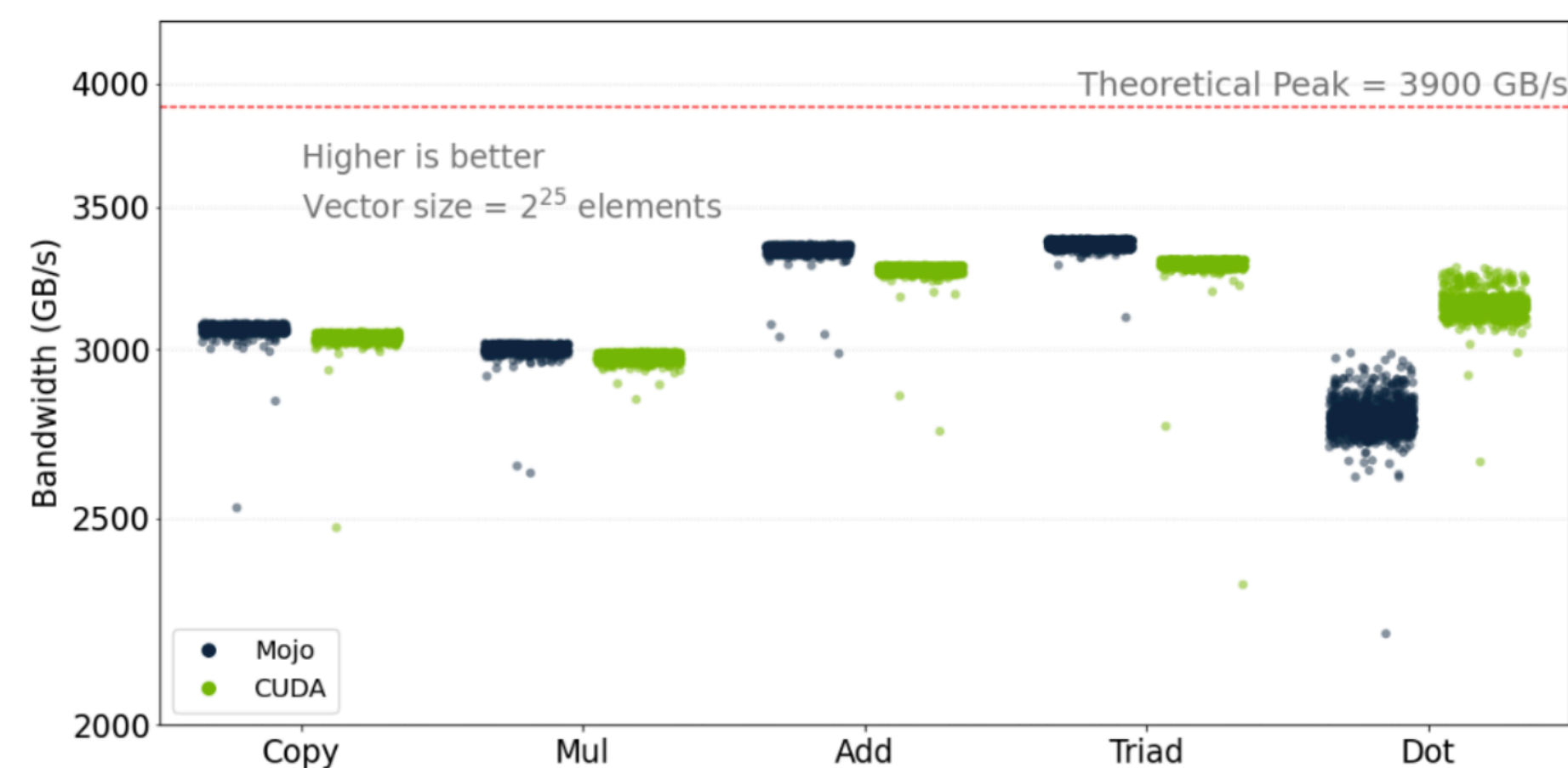
Outlook

- Near term, fixable
 - fast-math, AMD atomics
- Bigger lifts
 - multi-node execution
- Fall 2026
 - Modular has committed to open-sourcing Mojo
- MLIR is already used a lot. The question is will Mojo be used as well?

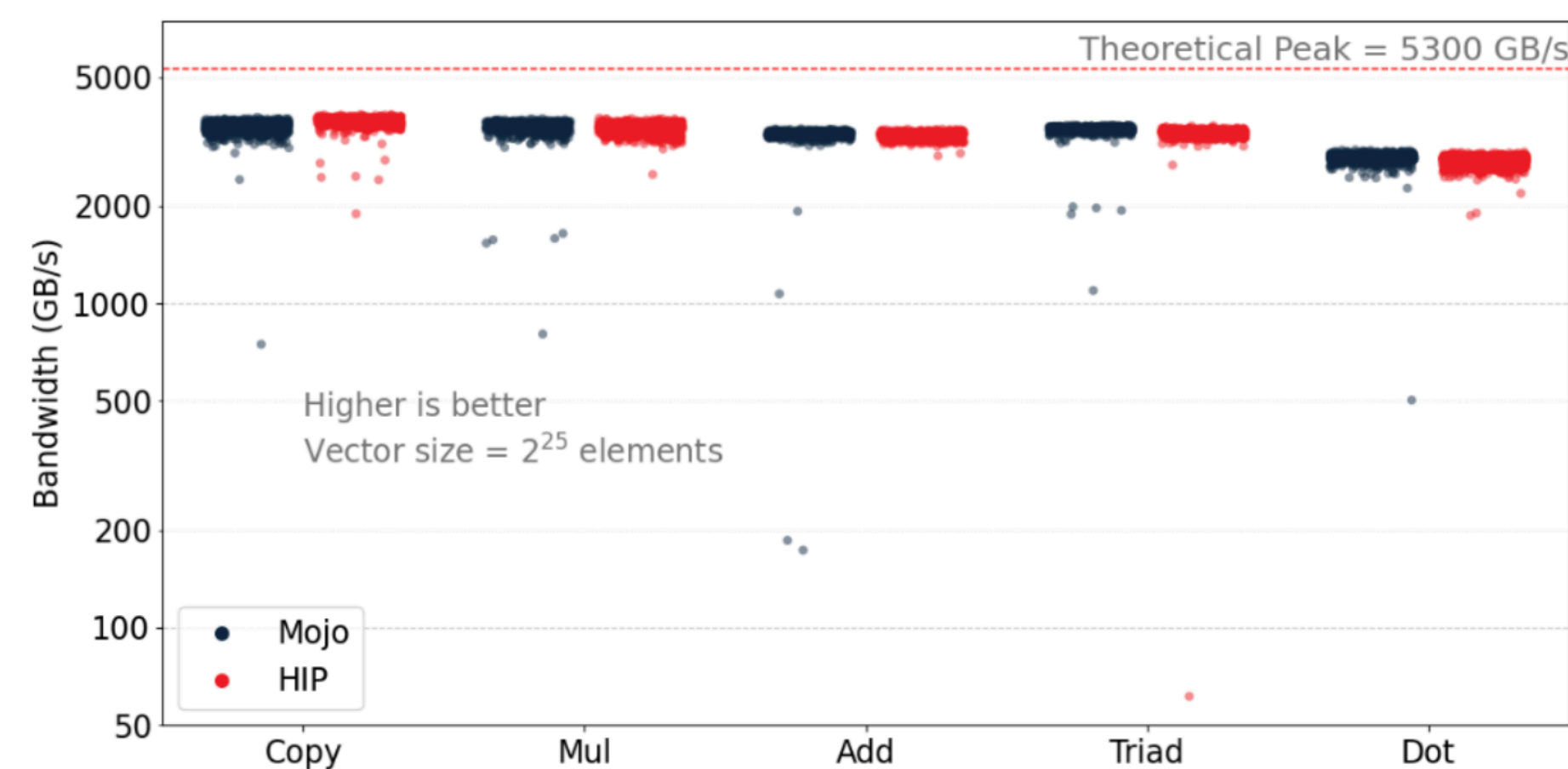
Summary

- Problems: two-language split and vendor lock-in
- MLIR, multi-level IR with dialects and progressive lowering. Hardware-specific code only in the last step. Portability becomes a compiler property
- Mojo's solution, a language built as MLIR dialects
- Near vendor parity memory-bound ($\Phi \sim 0.9$), real gaps compute-bound
- Verdict, right architecture, unfinished engineering

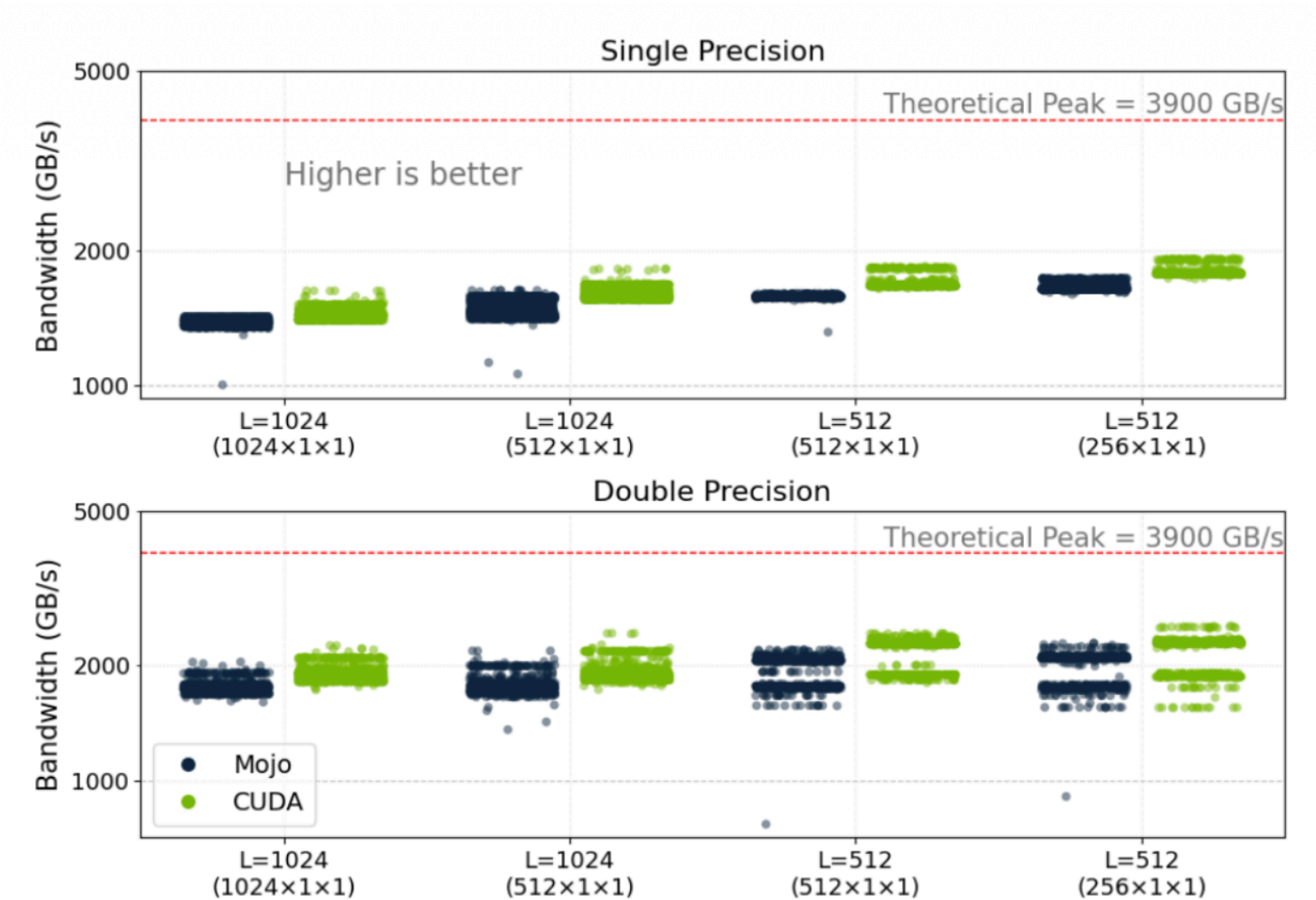
Extra: Babelstream and seven-point results



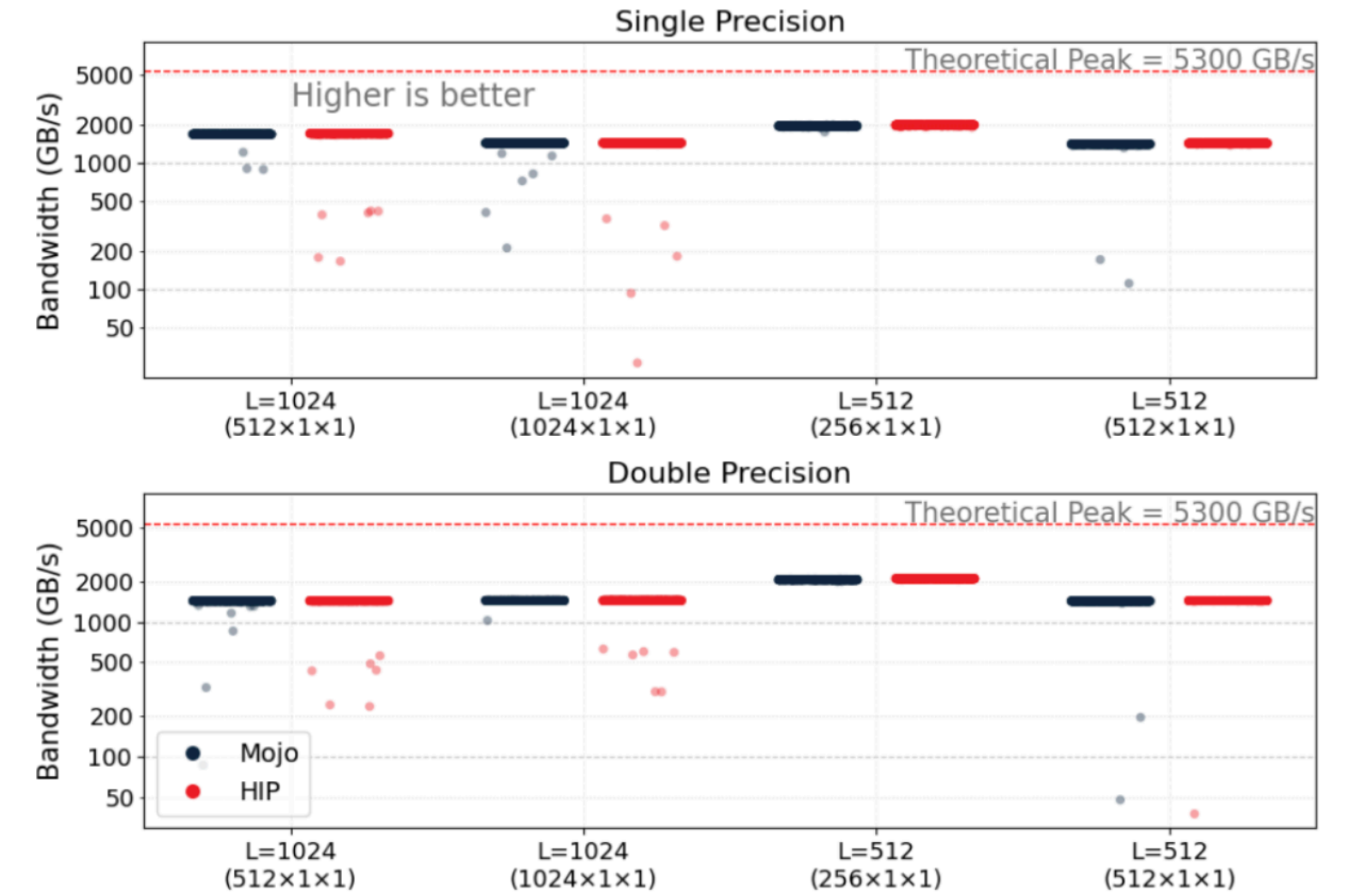
(a) H100



(b) MI300A



(a) H100



(b) MI300A